

Circuit Analysis (Lab # 6)

Experiment 6: VERIFICATION OF KIRCHHOFFS LAWS

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Date: 4/26/2026

Grade and Signature:

CLO2	Use with identification the circuit components, breadboards, multimeters, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
CLO3				
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objective

To verify Kirchhoff's current law and Kirchhoff's voltage law for the given Circuit.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	330ohm, 220ohm 1kohm	3

Description

KCL: The algebraic sum of the currents meeting at a node is equal to zero.

KVL: In any closed path / mesh, the algebraic sum of all the voltages is zero.

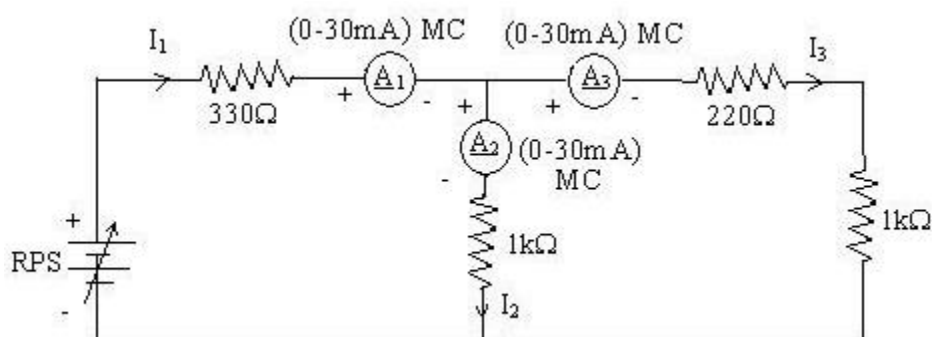
Precautions:

1. Voltage control knob should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position.

Procedure for KCL:

1. Give the connections as per the circuit diagram.
2. Set a value in RPS.
3. Note down the corresponding readings using multimeter
4. Repeat the same for different voltages

Circuit - KCL



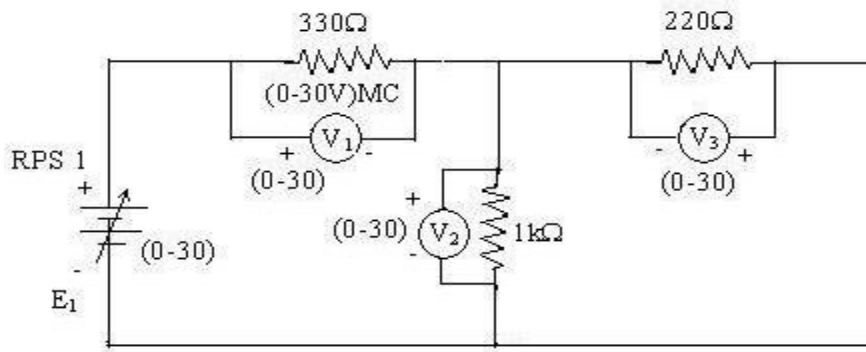
KCL - Practical Values:

Sl. No.	Voltage E Volts	Current			$I_1 = I_2 + I_3$ mA
		I_1 mA	I_2 mA	I_3 mA	
1	5	5.68	3.12	2.16	5.68
2	15	17.5	9.37	7.68	17.5
3	25	28.4	15.62	12.8	28.42

Procedure for KVL:

1. Give the connections as per the circuit diagram.
2. Set a value in RPS.
3. Note all the voltage reading
4. Repeat the same for different voltages

Circuit - KVL



KVL - Practical Values

Sl. No.	RPS	Voltage			$E1 = V1 + V2$
	E1	V1	V2	V3	
	V	V	V	V	V
1	5	3.23	1.77	1.77	5.00
2	10	6.47	3.53	3.53	10.00
3	15	9.70	5.3	5.3	15.00

Result:

Thus, Kirchhoff's voltage law and Kirchhoff's current law are verified practically by performing this experiment.

Post Lab Questions

1. Define Kirchhoff's current law.

Kirchhoff's Current Law (KCL) states that the total current entering a node or junction in a circuit is always equal to the total current leaving it. This is based on the principle of conservation of charge charge cannot accumulate at a node. Mathematically, the algebraic sum of all currents at any node equals zero ($\sum I = 0$).

2. Define Kirchhoff's voltage law.

Kirchhoff's Voltage Law (KVL) states that the algebraic sum of all voltages around any closed loop in a circuit is always equal to zero. This is based on the principle of conservation of energy, energy supplied by the source equals the energy consumed by the resistors. Mathematically, the sum of all voltage drops and rises in a closed loop equals zero ($\sum V = 0$).

3. What are the applications of regulated power supply?

A regulated power supply is used in electronic circuits and laboratory experiments to provide a stable, constant voltage regardless of load variations. It is widely used in powering microcontrollers, amplifiers, communication devices, and measuring instruments. It is also essential in battery charging systems, medical equipment, and industrial control systems where a steady voltage is critical for proper operation.

4. What are the limitations of KCL?

KCL assumes that the total current entering a junction equals the current leaving it, which holds true only for lumped circuit elements at low frequencies. At high frequencies, the effect of parasitic capacitance between conductors can cause current to leak, violating KCL. It also does not account for electromagnetic radiation effects in high-speed circuits, making it less accurate for microwave and RF circuit analysis.
